Problem 4

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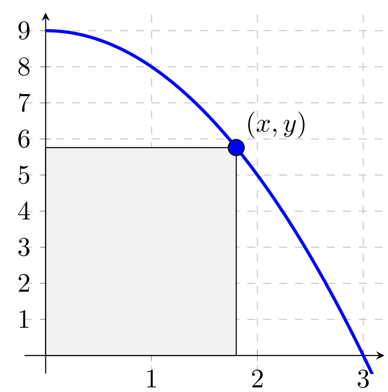
Problem 1 *A **rectangle in the first quadrant** is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x-axis and a line segment horizontally leftward to the y-axis, as in the picture below. Denote the **length of the base** (along the x-axis) of the rectangle by x, the **height of the rectangle** (along the y-axis) by y (in meters), and the **PERIMETER** of the rectangle by P. When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt}\right]_{x=2m}$ **by performing the steps below**.*



(a) ***Find a formula*** *for the height, y, of the rectangle as a function of x. (This is denoted as $y(x)$.)*

Explanation. The height of the rectangle is the y-coordinate of the point in the upper-right corner. Since this point lies on the graph of $f(x) = 9 - x^2$, we know $y = 9 - x^2$.

(b) ***Find the value*** *of* $\left[\frac{dx}{dt}\right]_{x=2m}$.

Explanation. Since $y = 9 - x^2$, we know $\frac{dy}{dt} = -2x\frac{dx}{dt}$. Solving for $\frac{dx}{dt}$

gives:

$$\begin{aligned}
\frac{dx}{dt} &= -\frac{\frac{dy}{dt}}{2x} \\
\left[\frac{dx}{dt}\right]_{x=2m} &= -\frac{\left[\frac{dy}{dt}\right]_{x=2m}}{2(2)} \\
&= -\frac{(-1/5)}{4} \\
&= \frac{1}{20}
\end{aligned}$$

(c) ***Find a formula*** *for the perimeter, P, of the rectangle as a function of x. (This is denoted as $P(x)$.)*

Explanation.

$$\begin{aligned}
P(x) &= 2x + 2y \\
&= 2x + 2(9 - x^2) \\
&= -2x^2 + 2x + 18
\end{aligned}$$

(d) ***Find the value*** *of* $\left[\frac{dP}{dt}\right]_{x=2m}$.

Explanation.

$$\begin{aligned}
P(x) &= -2x^2 + 2x + 18 \\
\frac{dP}{dt} &= (-4x + 2)\frac{dx}{dt}
\end{aligned}$$

$$\begin{aligned}
eval\frac{dP}{dt}_{x=2m} &= \left[(-4x + 2)\frac{dx}{dt}\right]_{x=2m} \\
&= (-4(2) + 2)\left(\frac{1}{20}\right) \\
&= -\frac{3}{10}
\end{aligned}$$

(e) ***Write a sentence to explain*** *what the value found in (d) means about the rectangle. (Don't forget UNITS.)*

Explanation. At the instant when $x = 2$ m, the perimeter of the rectangle is shrinking at the rate of $\frac{3}{10}$ m/s.